PROBLEM10:

PROBLEM STATEMENT:

Campus Event Financial Planning: Calculate venue, equipment, catering, transport, and certificate costs; determine participant contributions, sponsorship shortfalls, and the minimum registration count needed to break even.

OBJECTIVE:

To develop a Python program that calculates the total campus event expenses, including venue, equipment, catering, transport, and certificates. The program determines participant contributions, sponsorship shortfalls, and the minimum number of registrations required to break even.

PROGRAM:

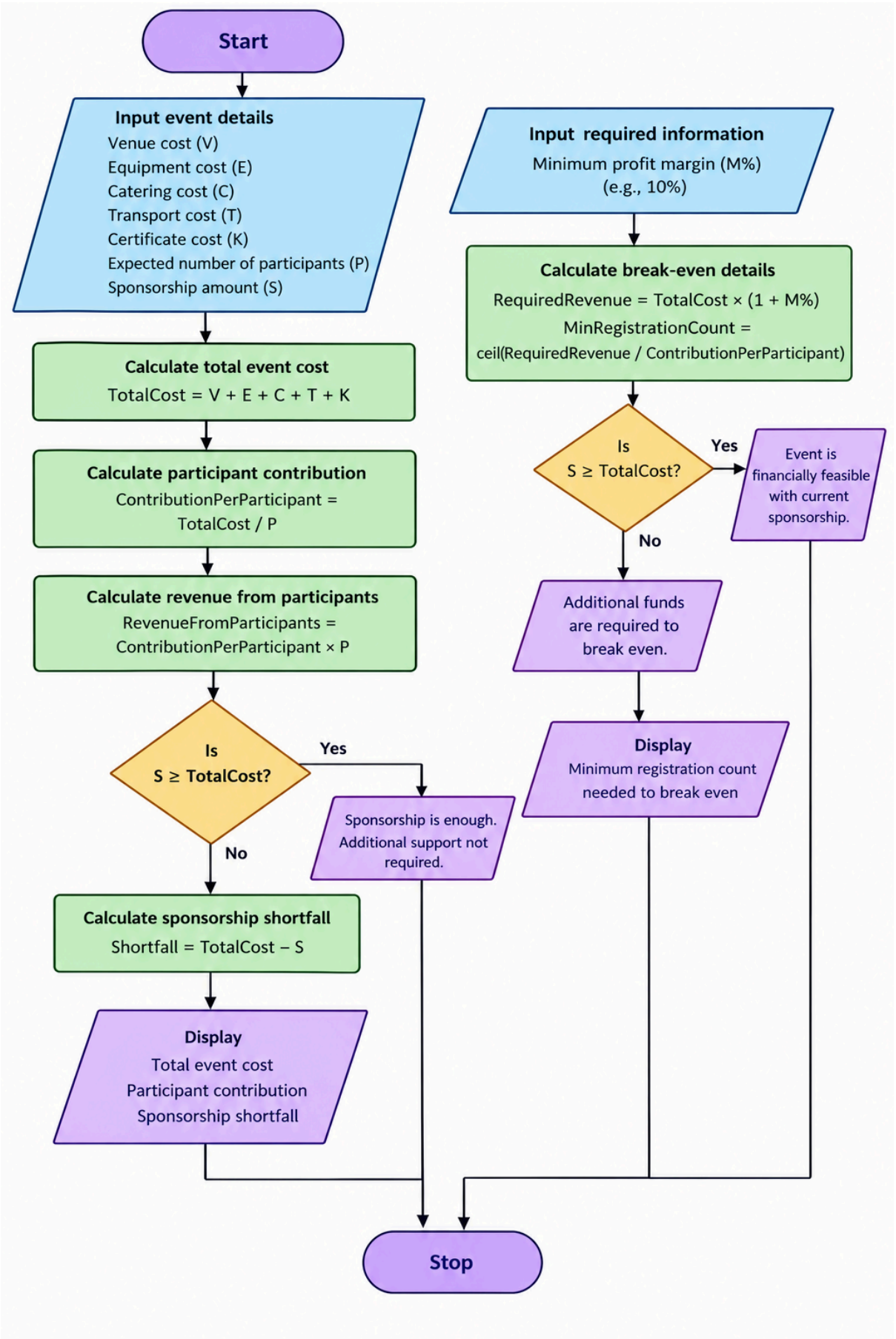
Campus Event Financial Planningvenue = float(input("Enter venue cost: "))equipment = float(input("Enter equipment cost: "))catering_per_person = float(input("Enter catering cost per person: "))transport = float(input("Enter transport cost: "))certificate_per_person = float(input("Enter certificate cost per person: "))sponsorship = float(input("Enter sponsorship amount: "))registration_fee = float(input("Enter registration fee per person: "))participants = int(input("Enter expected participants: "))# Calculate costsfixed_cost = venue + equipment + transportvariable_cost = catering_per_person + certificate_per_persontotal_variable = variable_cost * participantstotal_cost = fixed_cost + total_variableparticipant_contribution = registration_fee * participantstotal_funds = participant_contribution + sponsorshipshortfall = max(0, total_cost - total_funds)surplus = max(0, total_funds - total_cost)# Calculate minimum registrations to break evenif registration_fee > variable_cost: break_even = max(0,   int((fixed_cost - sponsorship) /   (registration_fee - variable_cost)) + 1   if (fixed_cost - sponsorship) /   (registration_fee - variable_cost) > 0   else 0   )else:

ALGORITHM:

1. Start.
2. Input venue, equipment, transport, catering, and certificate costs.
3. Input sponsorship amount, registration fee, and expected participants.
4. Calculate fixed costs by adding venue, equipment, and transport costs.
5. Calculate variable cost per participant by adding catering and certificate costs.
6. Calculate total event cost.
7. Calculate participant contributions and total available funds.
8. Determine any funding shortfall or surplus.
9. Calculate the minimum registrations required to break even.

10. Display the financial report.
11. Stop.

FLOWCHART:


Start
Input event details
Venue cost (V)
Equipment cost (E)
Catering cost (C)
Transport cost (T)
Certificate cost (K)
Expected number of participants (P)
Sponsorship amount (S)
Calculate total event cost
TotalCost = V + E + C + T + K
Calculate participant contribution
ContributionPerParticipant = TotalCost / P
Calculate revenue from participants
RevenueFromParticipants = ContributionPerParticipant × P
Is S ≥ TotalCost?
Yes
Sponsorship is enough. Additional support not required.
No
Calculate sponsorship shortfall
Shortfall = TotalCost − S
Display
Total event cost
Participant contribution
Sponsorship shortfall
Input required information
Minimum profit margin (M%) (e.g., 10%)
Calculate break-even details
RequiredRevenue = TotalCost × (1 + M%)
MinRegistrationCount = ceil(RequiredRevenue / ContributionPerParticipant)
Is S ≥ TotalCost?
Yes
Event is financially feasible with current sponsorship.
No
Additional funds are required to break even.
Display
Minimum registration count needed to break even
Stop

PSEUDOCODE:

```
BEGIN
   INPUT venue, equipment, transport
   INPUT catering_per_person, certificate_per_person
   INPUT sponsorship, registration_fee, participants

   fixed_cost = venue + equipment + transport
   variable_cost = catering_per_person + certificate_per_person

   total_cost = fixed_cost + variable_cost * participants
   participant_contribution = registration_fee * participants
   total_funds = participant_contribution + sponsorship

   shortfall = MAX(0, total_cost - total_funds)
   surplus = MAX(0, total_funds - total_cost)

   IF registration_fee > variable_cost THEN
      break_even = MAX(0, CEILING(
         (fixed_cost - sponsorship) /
         (registration_fee - variable_cost)
      ))
   ELSE
      break_even = NOT POSSIBLE
   END IF

   DISPLAY total_cost, participant_contribution
   DISPLAY shortfall, surplus, break_even
END
```

OUTPUT:

Example inputs:

- Venue: ₹10,000
- Equipment: ₹5,000
- Catering: ₹200 per person
- Transport: ₹3,000
- Certificates: ₹50 per person
- Sponsorship: ₹8,000
- Registration fee: ₹400 per person
- Expected participants: 100

--- Campus Event Financial Report ---

Fixed Cost: 18000.0
Variable Cost per Person: 250.0

Total Event Cost: 43000.0
Participant Contributions: 40000.0
Sponsorship: 8000.0
Sponsorship/Funding Shortfall: 0
Surplus: 5000.0
Minimum Registrations to Break Even: 0

The event costs ₹43,000, while available funds total ₹48,000. Therefore, there is a surplus of ₹5,000.